

EEAIC - PLL Exercise II

Alessandro Vegni & Giulio Guerrieri

Design of ring VCO

Q2.1)

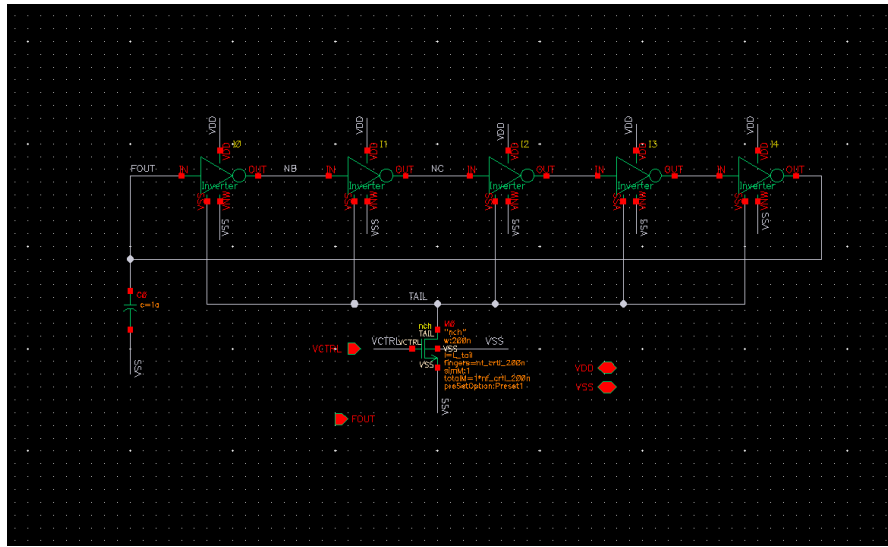


Figure 1: 5-Stage Ring Oscillator with Tail Current Control (M_0)

Analysis of M_0

The NMOS transistor M_0 of Figure 1 acts as a voltage-controlled current source. Its primary purpose is to control the charging and discharging current of the inverter stages, enabling frequency and power consumption tuning of the ring oscillator.

The oscillation frequency of a 5-stage ring oscillator is governed by the propagation delay (τ_d) of each inverter stage, given by:

$$f_{osc} = \frac{1}{2 \times 5 \times \tau_d} = \frac{1}{10\tau_d}$$

The propagation delay depends heavily on the rate at which the load capacitance can be discharged through the NMOS network. By varying the gate control voltage (V_{ctrl}) of

M_0 , the tail current is modulated, directly controlling τ_d and thus tuning the oscillation frequency.

To ensure a predictable, stable, and linear voltage-to-frequency relationship, M_0 must remain continuously in the saturation region throughout the entire oscillation cycle. When V_{ctrl} increases, I_{tail} increases, which speeds up the discharge time, reducing τ_d and increasing f_{osc} . Conversely, lowering V_{ctrl} starves the inverter stages of current, increasing τ_d and lowering f_{osc} .

Output Voltage Swing

Because the source terminals of the inverter NMOS transistors are no longer tied directly to V_{SS} , they sit at a virtual ground potential equal to $V_{DS,M0}$.

Assuming M_0 must remain in saturation to function as a proper current source, its minimum voltage drop is its overdrive voltage:

$$V_{DS,M0} \geq V_{ctrl} - V_{th,M0} = V_{DS,sat}$$

Consequently, the output voltage cannot pull all the way down to absolute V_{SS} . The expected voltage swing at the output of the inverters is reduced and ranges from:

$$V_{out,min} = V_{DS,sat} \quad \text{to} \quad V_{out,max} = V_{DD}$$

Therefore, the maximum output peak-to-peak voltage swing is:

$$V_{swing,max} = V_{DD} - V_{DS,sat}$$

Actual output voltage swing is:

$$V_{swing} = V_{DD} - V_{DS,M0}$$

Q2.2) Selection of Component Values (C_0 and R_0)

The network formed by C_0 and R_0 acts as a first-order RC high-pass filter. The sizing of these components is governed by two main criteria: frequency response and capacitive loading.

1. High-Pass Corner Frequency The high-pass pole frequency (f_p) must be placed significantly below the minimum operating frequency of the VCO ($f_{osc,min}$) to prevent attenuation and phase shift of the oscillator signal:

$$f_p = \frac{1}{2\pi R_0 C_0} \ll f_{osc,min} \implies R_0 C_0 \gg \frac{1}{2\pi f_{osc,min}}$$

2. Sizing Strategy

- **Feedback Resistance (R_0):** R_0 is chosen to be very large to minimize the AC current loaded from the output back to the input, preventing a reduction in the inverter's open-loop gain. Additionally, a larger R_0 allows for a smaller, more area-efficient integrated capacitor C_0 .
- **Coupling Capacitance (C_0):** Once R_0 is fixed, C_0 is chosen to satisfy the high-pass corner frequency condition. Furthermore, C_0 must be significantly larger than the total input parasitic capacitance ($C_{in,inv} \approx C_{ggn} + C_{gpp}$) of the restoring inverter to avoid signal attenuation via the capacitive voltage divider:

$$V_{in,inv} = V_{vco} \left(\frac{C_0}{C_0 + C_{in,inv}} \right)$$

Therefore, the design rule $C_0 \gg C_{in,inv}$ must be maintained.

Q2.3) Figure of Merit for VCO

The Figure of Merit of a Voltage-Controlled Oscillator (VCO) is a standardized performance metric used to evaluate and compare the intrinsic quality of different oscillator topologies. Because phase noise can always be artificially lowered at the expense of burning more power, or varies inherently based on the carrier frequency (f_{osc}) and the selected offset frequency (Δf), a raw phase noise measurement is insufficient for cross-architecture comparison.

The FoM normalizes these design trade-offs into a single figure. It represents the phase noise performance normalized to an operating frequency of 1 GHz, an offset of 1 MHz, and a power consumption of 1 mW. A lower (more negative) FoM represents a more power-efficient design with superior noise performance.

To calculate the FoM, we first define the Single-Sideband (SSB) phase noise, $L(\Delta f)$, which is the ratio of the noise power in a 1 Hz bandwidth at a specific frequency offset $\Delta f = f - f_0$ from the carrier to the total carrier power:

$$L(\Delta f) = 10 \log \left(\frac{P_{noise}(1 \text{ Hz at } f)}{P_{carrier}} \right) \quad [\text{dBc/Hz}]$$

Using the SSB phase noise, the FoM is calculated via the following standardized formula:

$$\text{FoM} = L(\Delta f) - 20 \log \left(\frac{f_{osc}}{\Delta f} \right) + 10 \log \left(\frac{P_{VCO}}{1 \text{ mW}} \right) \quad [\text{dB}]$$

Where f_{osc} is the fundamental oscillation (carrier) frequency and P_{VCO} is the DC power consumption of the oscillator core.

The FoM remains constant regardless of the chosen offset Δf , provided that the measurement is taken within the $1/f^2$ (white-noise-dominated) region of the phase noise spectrum where $L(\Delta f)$ rolls off at a rate of -20 dB/dec .

According to foundational limits established in literature (Navid et al., 2005), the absolute theoretical limit for Ring Oscillator is bounded near -165 dB.

A3.1) Sizing of Ring VCO

To satisfy the performance metrics a systematic, iterative sizing methodology was deployed.

1. Power and Core Current Initialization

The total target power consumption is bounded at 2 mW. Given a standard nominal supply voltage of $V_{DD} = 1.2$ V, the maximum allowable total current drawn from the supply is:

$$I_{\text{total}} = \frac{P_{\text{VCO}}}{V_{DD}} = \frac{2 \text{ mW}}{1.2 \text{ V}} \approx 1.67 \text{ mA}$$

Allocating roughly 1.4 mA to the oscillator core (leaving the remainder for the buffer stages), the initial design phase focused on finding the appropriate aspect ratio (W/L) for the tail NMOS current source (M_0) to provide this current under a fixed gate control voltage of $V_{\text{CTRL}} = 0.75$ V while guaranteeing operation strictly within the saturation region.

2. Frequency Tuning and Inverter Core Sizing

Once the baseline core current was established, the widths of the 5-stage ring inverters were tuned to hit the target oscillation frequency of 2 GHz. To achieve symmetric rise and fall times and establish a balanced inversion profile, the PMOS transistors were sized with a width twice as large as the NMOS transistors.

As stated before, the fundamental frequency of an N -stage ring oscillator is governed by the propagation delay per stage (τ_d), which is heavily dependent on the inverter's load. Thus frequency tuning was carried out considering variations on capacitive-loading of the devices:

- If the observed frequency was too high, the inverter widths were proportionally increased to intentionally introduce larger self-loading gate and drain parasitic capacitances, thereby slowing down the stages.
- Conversely, if the frequency fell below 1.9 GHz, the inverter sizes were decreased to reduce parasitic capacitive loading and accelerate the switching transitions.

Because modifying the core inverter sizes inherently changes the dynamic switching current and alters the voltage headroom, an iterative optimization loop was executed: adjusting the aspect ratio of the tail current source M_0 to restore the power budget, followed by fine-tuning the inverter dimensions to lock down the 2 GHz frequency specification.

3. Transient Swing Restoration

The AC-coupled buffer, sized almost the same as the core components, proved robust enough to fully restore the degraded internal core swing. As illustrated in Figure 2, the final output signal (OUT) demonstrates a clean, symmetric digital swing from V_{SS} (0 V) to V_{DD} (1.2 V).

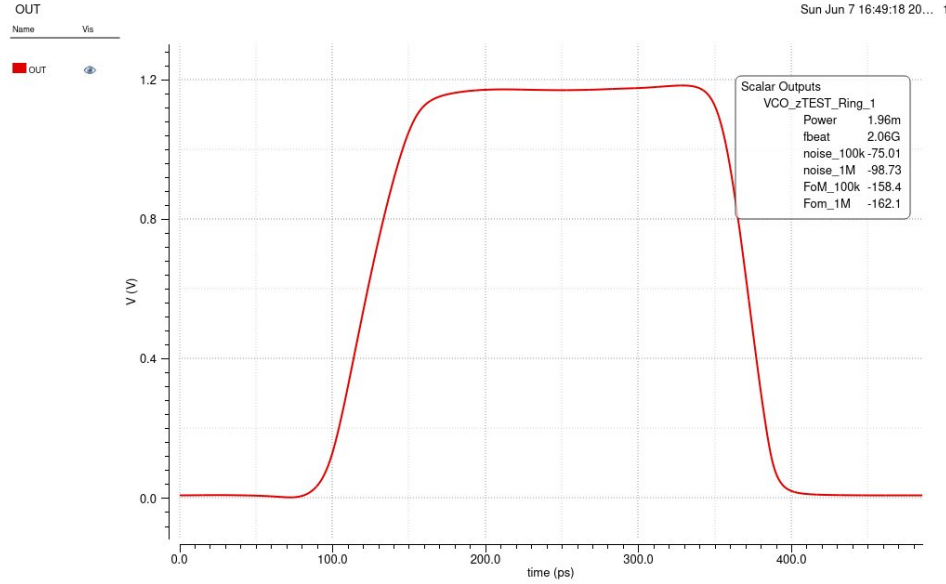


Figure 2: Output Full swing and obtained Design Metrics

4. Figure of Merit Optimization

Initial Periodic Noise evaluations indicated that the phase noise performance was inadequate, leading to a degraded Figure of Merit. A noise summary analysis revealed that the tail transistor M_0 was overwhelmingly the dominant noise contributor to the circuit ($\approx 90\%$ contribution), primarily driven by its low-frequency flicker ($1/f$) noise component upconverted near the carrier.

To suppress this flicker noise without altering the locked current (1.4 mA) or shifting the frequency-setting parameters, proportional area scaling was implemented. The intrinsic flicker noise voltage spectral density is inversely proportional to the active gate area:

$$S_{V_{th},1/f} = \frac{K_f}{C_{ox} \cdot W \cdot L \cdot f} \quad (1)$$

Both the width (W_{M0}) and length (L_{M0}) of the tail current source were scaled up by almost the same scaling factor. This design choice preserved the established current drivem while significantly expanding the physical area.

This optimization dramatically reduced the contribution of flicker noise. Following successive scaling iterations, the definitive dimensions of the transistor were finalized, as detailed in Figure 6.

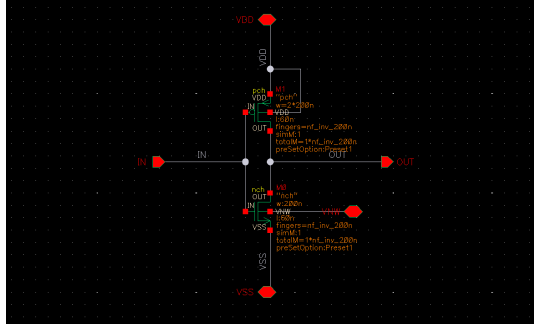


Figure 3: Core inverter schematic

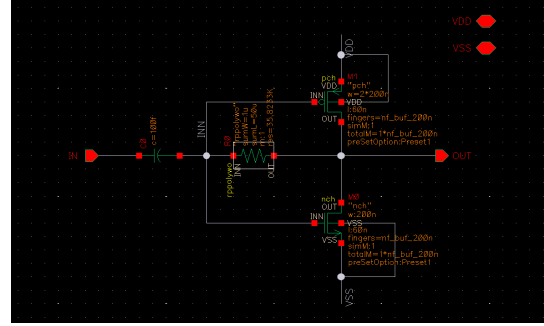


Figure 4: RingBuffer

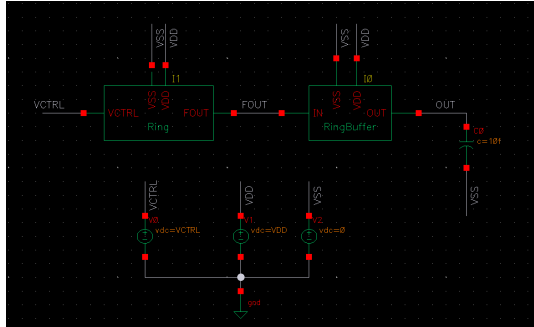


Figure 5: Ring testbench setup

Design Variables	
VCTRL	0.75
VDD	1.2
approx_freq	2G
nf_buf_200n	130
nf_inv_200n	140
nf_crtl_200n	45*15
L_tail	60n*45

Figure 6: Final values of Design variables

The final, post-optimization Pnoise measurements successfully respected all target metrics. As shown in Figure 2 the calculated Figures of Merit achieved are:

- **At 100 kHz offset:** FoM = -158.4 dBc/Hz
- **At 1 MHz offset:** FoM = -162.1 dBc/Hz

A3.2) Identification and Mitigation of Major Noise Contributors

To identify the primary noise mechanisms limiting the oscillator's performance, a spot noise summary was generated at the two target offset frequencies of interest: 100 kHz and 1 MHz. As illustrated in Figure 7, the data clearly indicates that the tail current source transistor (M_0) is the absolute dominant noise bottleneck in the entire system, through two distinct physical phenomena depending on the frequency band: flicker noise (f_n) dominating close to the carrier at 100 kHz, and channel thermal noise (i_d) dictating the noise floor further out at 1 MHz.

Device	Param	Noise Contribution	% Of Total
/I1/M0	fn	3.44175e-09	39.06
/I1/M0	id	2.26608e-09	25.71
/I1/I1/M0	fn	3.01424e-10	3.42
Spot Noise Summary (in V ² /Hz) at 100K Hz Sorted By Noise Contributors			
Total Summarized Noise = 8.81247e-09			
No input referred noise available			
Total noise by type:			
Flicker: 0.00000E+00			
Thermal: 2.26608E-09			
Shot: 0.00000E+00			
The above noise summary info is for pnoise_pm data			
Device	Param	Noise Contribution	% Of Total
/I1/M0	id	2.14707e-11	61.17
/I1/M0	fn	3.55751e-12	10.14
/I1/I4/M1	id	9.86423e-13	2.81
Spot Noise Summary (in V ² /Hz) at 1M Hz Sorted By Noise Contributors			
Total Summarized Noise = 3.51008e-11			
No input referred noise available			
Total noise by type:			
Flicker: 0.00000E+00			
Thermal: 2.24571E-11			
Shot: 0.00000E+00			
The above noise summary info is for pnoise_pm data			

Figure 7: Noise summary at 100 kHz and 1 MHz

At the 100 kHz offset low-frequency $1/f$ noise of M_0 is aggressively upconverted near the carrier frequency. To address this major bottleneck, a proportional area scaling strategy was implemented by significantly increasing both the width (W) and length (L) of the tail transistor, since flicker noise power is inversely proportional to $W \cdot L$. Scaling both dimensions upward preserved the required current drive to meet target power consumption.

At the higher offset of 1 MHz channel thermal noise of M_0 is the primary contributor. To mitigate this high-frequency contribution under a fixed current budget, the transistor's channel length L was intentionally increased a bit more than the width W . This deliberate reduction in the overall W/L ratio effectively lowered the transconductance according to the relationship $g_m \approx \sqrt{2\mu_n C_{ox}(W/L)I_{tail}}$, dampening the injected thermal noise power.

A3.3) Phase Noise Characteristics and Region Identification

The simulated phase noise spectrum is plotted across a offset frequency range from 1 kHz to 1 GHz, as shown in Figure 8.

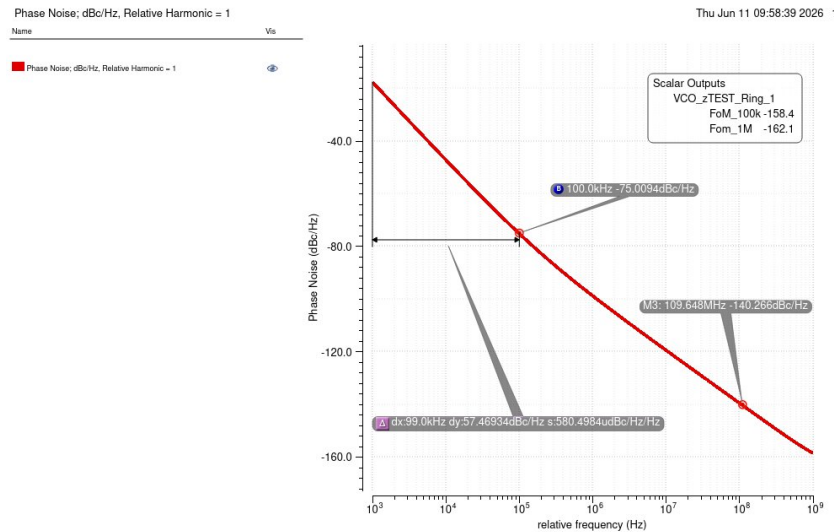


Figure 8: Simulated phase noise spectrum as a function of relative offset frequency from the carrier.

The phase noise curve exhibits distinct operational regions characterized by different slopes, reflecting the underlying physical noise mechanisms upconverted from the devices:

1. The $1/f^3$ Close-In Region (Flicker Phase Noise)

At low frequency offsets — extending up to approximately 100 kHz — the phase noise drop-off follows a steep slope of roughly -30 dB/dec.

This region is dictated by the intrinsic flicker ($1/f$) noise of the transistors, primarily originating from the tail current source M_0 . The nonlinear switching behavior of the ring oscillator modulates and upconverts this low-frequency baseband flicker noise around the 2 GHz carrier, adding an extra $1/f$ dependency to the standard oscillator phase noise response and generating the $1/f^3$ characteristic.

2. The $1/f^2$ Region (White Phase Noise)

For frequency offsets spanning roughly from 100 kHz past 10 MHz, the curve transitions into a classic -20 dB/dec slope.

This region is dominated by the white thermal noise of the circuit components. White noise possesses a flat spectral density across all frequencies; when processed through the ideal integration function of the ring oscillator, it accumulates a standard $1/f^2$ frequency roll-off.

3. Absence of the Wideband White Noise Floor

As the offset frequency scales toward the maximum simulated limit of 1 GHz, the phase noise curve does not transition into a horizontal, flat thermal noise floor; instead, it

maintains a strict -20 dB/dec trajectory.

In textbook oscillator profiles, far-out phase noise eventually flattens due to the unshaped, additive white thermal noise floor of the output buffers. However, because a ring oscillator lacks a high-Q resonant tank, its core thermal noise filtering is relatively weak. This causes the integrated $1/f^2$ noise from the switching core to remain highly dominant over a massive frequency span.

As verified by the plot, the core noise drops to approximately -160 dBc/Hz at a 1 GHz offset. Since the wideband thermal noise floor of the output buffers typically resides at a lower level (between -165 dBc/Hz and -175 dBc/Hz), the crossover point where the curve flattens lies beyond the simulated 1 GHz window. Therefore, the thermal noise of the core remains the definitive bottleneck across the entire high-frequency offset spectrum shown.

Design of LC VCO

A4.1) Component Functionality in Class-B VCO

- **Center-Tapped Symmetric Inductor:** The center tap between the two symmetric halves of the inductor is directly tied to the DC supply voltage V_{DD} , establishing the bias path for the cross-coupled pair. As a result, the output common-mode voltage is effectively fixed at V_{DD} , thus preventing it from being unintentionally modulated by tail current noise. Additionally, implementing the structure as a single symmetric interleaved spiral utilizes mutual magnetic coupling between the two halves to achieve a significantly higher Quality Factor and a reduced silicon area footprint, leading to a reduction in the oscillator's phase noise.
- **Parallel MOS Varactors:** Continuous frequency tuning is provided by MOS varactors placed in parallel with the LC tanks. Because the inductor's center tap holds the output CM level at V_{DD} , the gates of the varactors sit at an average voltage of V_{DD} . As the voltage applied to the source/drain terminals varies, the varactor's gate-source voltage changes monotonically, steering the capacitance from a minimum (C_{min}) to a maximum (C_{max}). The resulting oscillation frequency is defined by $\omega_{osc} = 1/\sqrt{LC(V)}$.
- **NMOS Cross-Coupled Pair:** The NMOS cross-coupled transistors serve as the active core of the oscillator by a negative resistance to compensate for the equivalent parallel resistance R_p of the lossy LC tank. In resistance terms, a stronger parallel resistance is a smaller ohmic value ($1/g_m < R_p$). Therefore, the loop gain of the circuit must reach or exceed unity to successfully start and sustain oscillation, requiring a minimum transconductance dictated by the startup condition $g_m R_p \geq 1$.
- **Tail Transistor and Current Mirror:** The tail transistor defines the total bias current I_{SS} of the oscillator, providing a much more robustly defined current than topologies without a tail source. Nonetheless, the tail current source and its associated reference mirror introduce additional noise penalties.

A4.2) Derivation of Tank Resistance R_p and Inductance

In steady-state oscillations, the cross-coupled pair (XCP) transistors alternate between being ON and OFF, acting as a hard current-steering switch. Consequently, the current waveform flowing into a single output node over time resembles a perfect square wave toggling between 0 and I_{SS} with a 50% duty cycle.

Using a Fourier series expansion, this current waveform $i(t)$ can be mathematically expressed as:

$$i(t) = \frac{I_{SS}}{2} + \frac{2I_{SS}}{\pi} \cos(\omega_0 t) - \frac{2I_{SS}}{3\pi} \cos(3\omega_0 t) + \frac{2I_{SS}}{5\pi} \cos(5\omega_0 t) - \dots$$

The resonant LC tank acts as a frequency-selective filter with the following behavior:

- **At DC ($\omega = 0$):** The inductor L acts as a perfect short circuit to the supply rail V_{DD} , yielding zero AC voltage drop.
- **At Higher Harmonics ($3\omega_0, 5\omega_0, \dots$):** The tank capacitance C acts as a low-impedance path to ground, suppressing higher frequency current components.
- **At the Fundamental Resonant Frequency (ω_0):** The ideal reactive elements perfectly cancel each other out $Z_L \parallel Z_C = \infty$, leaving only the equivalent parallel tank resistance R_p to determine the total node impedance.

Therefore, the single-ended voltage swing V_{swing} is driven entirely by the fundamental AC current component ($I_1 = \frac{2}{\pi} I_{SS}$) interacting with R_p :

$$V_{swing} = \frac{2}{\pi} I_{SS} R_p$$

Rearranging this expression to solve for the required parallel resistance R_p to achieve a single-ended voltage swing of $V_{swing} = 0.6$ V with a tail current of $I_{SS} = 2$ mA:

$$R_p = \frac{\pi \cdot V_{swing}}{2 \cdot I_{SS}} \approx \mathbf{471.24 \, \Omega}$$

To calculate the total required tank inductance, we apply the definition of the Quality Factor Q for a parallel resonant tank model:

$$Q = \frac{R_p}{\omega_0 L}$$

Given an oscillation frequency of $f_0 = 2$ GHz and an inductor quality factor of $Q = 15$, isolating L yields:

$$L = \frac{R_p}{\omega_0 Q} = \mathbf{2.5 \, nH}$$

A4.3) Analysis of Voltage Swing Limits

In a cross-coupled oscillator, the oscillation amplitude ($V_{swing} = V_a$) must be rigorously balanced. Deviations toward either excessively large or insufficient amplitude can lead to significant degradation of circuit performance.

Case V_{swing} is Excessively Large

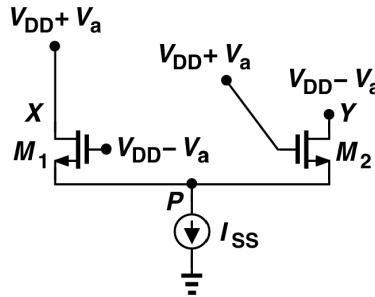


Figure 9: Voltage swings in cross-coupled oscillator

Referring to the peak of the differential oscillation cycle illustrated in *Figure 9* the circuit nodes experience asymmetric maximum voltage transitions, as shown in "RF Microelectronics" of Razavi.

If the swing becomes excessively large, the voltage differences across the transistors become extreme, pushing the devices toward their physical limits:

- **Triode Region Entry:** At the peak of the cycle, the conducting transistor (e.g. M_2) sees a very high gate voltage $V_{DD} + V_a$ and a very low drain voltage $V_{DD} - V_a$. Indeed, if V_a is sufficiently big then $V_{GD} = 2V_a > V_{th}$ pushing the transistor deep into the triode region, which lowers its effective transconductance. Additionally when in triode regime, the channel acts as a simple resistance connecting the LC tank directly to ground, ruining the Q-factor and thus the noise performance.
- **Device Stress:** The massive voltage differences across the gate-drain and gate-source terminals can exceed the safe limits of the oxide layer, accelerating long-term breakdown.
- **Varactor AM-to-PM Conversion:** An excessive voltage swing can lead to varactor AM-to-PM conversion. Under these conditions, any amplitude noise on the supply line dynamically modulates the average capacitance of the varactor, upconverting power supply fluctuations directly into phase noise.
- **Short-Channel Effects:** The OFF transistor (e.g. M_1) must withstand a massive drain-to-source voltage. In modern nanoscale nodes, this leads to disruptive leakage mechanisms:
 - **DIBL (Drain-Induced Barrier Lowering):** The high drain voltage lowers the internal barrier, causing unwanted subthreshold leakage.
 - **GIDL (Gate-Induced Drain Leakage):** High electric fields cause tunneling, introducing additional body leakage.

- **Punchthrough:** If the voltage gets high enough, the depletion regions of the source and drain touch each other. The gate completely loses control of the channel, leading to a massive subsurface current that can permanently destroy the device.

Case V_{swing} is Excessively Small

A very small voltage swing typically indicates that the circuit is operating with insufficient tail current I_{SS} or that the tank's parallel resistance R_p is heavily degraded. This creates two primary failure mechanisms:

- **Startup Failure:** In the saturation region, a transistor's transconductance is dictated by its bias current. If I_{SS} is too low, the initial small-signal g_m is critically weak. The loop gain fails to satisfy: $g_m R_p \geq 1$, meaning the oscillation simply never starts.
- **Incomplete Switching:** Because the peak-to-peak swing is small, the differential voltage applied to the cross-coupled gates is insufficient to act as a hard switch. The transistors fail to fully steer the tail current, meaning both devices remain in the active saturation region for a significant portion of the cycle.

A4.4) Length of the XCP Transistors

To ensure a robust start-up for the Class-B VCO, the cross-coupled pair must be sized to provide sufficient transconductance (g_m). Small-signal theory dictates that the theoretical minimum boundary for oscillation is:

$$g_m R_p \geq 1,$$

which corresponds to the exact point where the negative resistance generated by the cross-coupled pair compensates for the energy dissipated by the tank's parasitic parallel resistance R_p . However, this theoretical limit is only valid under small-signal conditions.

As the oscillation amplitude increases and large voltage swings develop, the transistors fully steer the tail current from one branch to the other. During portions of the oscillation cycle, the effective transconductance temporarily falls below $1/R_p$. Consequently, practical designs require a safety margin to guarantee reliable start-up and complete current steering throughout the entire oscillation cycle.

A common design guideline is to size the transconductance to be approximately three to four times larger than the theoretical minimum. Following this approach, the target transconductance was chosen as:

$$g_m = 3 \left(\frac{1}{R_p} \right) = 6.37 \text{ mA/V}.$$

Once the target transconductance was established, selecting the appropriate channel length L required balancing a delicate trade-off between noise performance and parasitic

capacitance. Instead of using the minimum channel length allowed by the technology $L_{\min}$, the transistors were designed with a channel length approximately two to three times larger than $L_{\min}$.

- **Too Short Channel Length:** Since flicker noise is inversely proportional to the transistor area, smaller devices exhibit higher flicker noise levels. In a VCO, this low-frequency noise is particularly harmful because it is upconverted around the carrier frequency, thereby degrading the noise performance of the oscillator.
- **Excessively Long Channel Length:** To maintain the necessary W/L ratio and achieve the target transconductance, an excessively large channel length requires a proportional increase in transistor width W leading to substantial parasitic capacitances. These parasitic capacitances load the LC resonator lowering the maximum achievable oscillation frequency and reducing the VCO tuning range.

A4.5) Capacitor Bank Design for the VCO

1) Total Capacitance C_{res}

Considering a target oscillation frequency of $f_{\text{target}} = 2 \text{ GHz}$ and a tank inductance of $L = 2.5 \text{ nH}$, the resonant capacitance required to achieve the desired center frequency can be obtained from the LC resonance condition as:

$$C_{\text{res}} = \frac{1}{(2\pi f_{\text{target}})^2 L} \approx 2.533 \text{ pF}$$

2) Tuning Range Capacitance ΔC

A tuning range of $\pm 5\%$ around the center frequency is specified, corresponding to a total frequency span of 200 MHz. Therefore, the oscillator must be tunable between:

$$f_{\min} = 1.9 \text{ GHz} \quad \text{and} \quad f_{\max} = 2.1 \text{ GHz}$$

The maximum capacitance is required at the minimum oscillation frequency and is given by:

$$C_{\max} \Big|_{f_{\min}=1.9 \text{ GHz}} = \frac{1}{(2\pi f_{\min})^2 L} \approx 2.807 \text{ pF}$$

Similarly, the minimum capacitance is required at the maximum oscillation frequency:

$$C_{\min} \Big|_{f_{\max}=2.1 \text{ GHz}} = \frac{1}{(2\pi f_{\max})^2 L} \approx 2.298 \text{ pF}$$

Consequently, the required tuning capacitance range is:

$$\Delta C_{\text{tune}} = C_{\max} - C_{\min} \approx 0.509 \text{ pF}$$

3) Sizing Switches MIM Capacitances

Sizing the MOS switch in the MIMCAP switches introduces a strict trade-off between parasitic resistance and capacitance:

- If the switch is sized too small: The device exhibits a high on-resistance (R_{on}). When the switch is closed to connect the MIM capacitors to the tank, this high channel resistance introduces a significant AC voltage drop. This loading effect severely attenuates the voltage swing (V_{swing}) across the LC tank, directly degrading the phase noise performance and threatening the oscillator's startup reliability. Additionally, a smaller channel area amplifies the device's intrinsic flicker noise contribution.
- If the switch is sized too large: The dominant bottleneck shifts from resistance to massive parasitic capacitances. When the switch is open (OFF), these unswitched parasitic capacitances remain permanently wired to the LC tank. This heavily compromises the tuning ratio of the VCO, narrowing the total available frequency sweep range.

A4.6) ClassB VCO Design

Capacitor Bank Architecture

Given the target VCO gain of $K_{VCO} = 5 \text{ MHz/V}$ operating over a control voltage (ΔV_{ctrl}) window of 400 mV, the maximum continuous frequency variation per state is 2.0 MHz. To cover the overall 200 MHz tunable frequency range, the absolute minimum number of discrete levels required is calculated as:

$$N_{levels} = \frac{200 \text{ MHz}}{2.0 \text{ MHz}} = 100$$

To achieve at least 100 discrete states, an N -bit binary control system is required such that $2^N \geq 100$ (implying a minimum of 7 bits). However, to guarantee continuous tuning and prevent tuning blind zones between adjacent bands, sufficient frequency overlap must be established. Consequently, the designed digital frequency step (Δf_{dig}) between consecutive levels must be strictly smaller than the 2.0 MHz analog tuning capability.

The architecture is structured as follows:

- **1 Analog-Controlled MOSCAP**
- **4 Binary-Weighted MOSCAPs:**
 - Bit 0 (LSB): $C = C_0$
 - Bit 1: $C = 2C_0$
 - Bit 2: $C = 4C_0$
 - Bit 3: $C = 8C_0$

• 3 Binary-Weighted MOMCAPs:

- Bit 4: $C = 16C_0$
- Bit 5: $C = 32C_0$
- Bit 6 (MSB): $C = 64C_0$

This deliberate sizing guarantees complete overlap between the smallest MSB capacitor and the largest LSB capacitor, inherently preventing holes in frequency tunability.

Unit Capacitance (C_0) Calculation

With a total of 128 levels (providing 127 digital steps) and a required total capacitance variation of $\Delta C = 0.51$ pF, the unit capacitance C_0 can be determined as follows:

$$127C_0 = 0.51 \text{ pF} \implies C_0 \approx 4 \text{ fF}$$

Because the architecture employs two capacitors in series at each stage, the physical value of each individual capacitor must be multiplied by two, yielding a baseline individual capacitance of approximately 8 fF. The MIMCAPs of the binary-weighted MOMCAPs and the varactors of the binary-weighted MOSCAPs are sized according to this baseline value and the aforementioned binary weighting system.

Analog-Controlled MOSCAP

The two transistors of the varactor, visible in *Figure 10*, were sized to target a VCO gain of $K_{\text{VCO}} = 5$ MHz/V, which determines the slope of the frequency versus the control voltage.

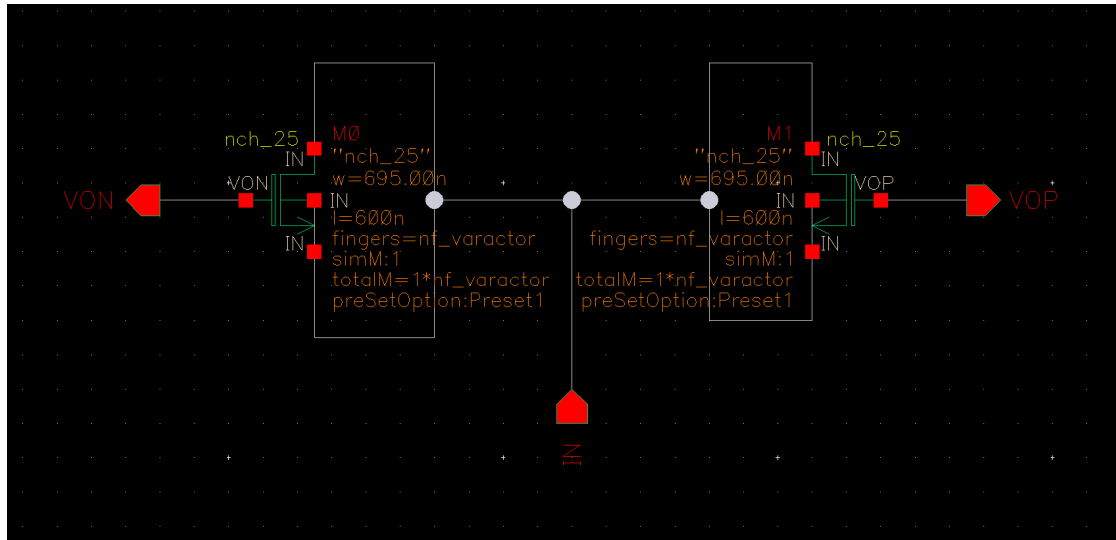


Figure 10: Schematic of the Analog-Controlled MOS Varactor

Based on the output constraints of the previously designed charge pump (CP), the control voltage (V_{ctrl}) operates within a linear range of 400 mV to 800 mV. Over this 0.4 V window, the analog varactor must provide a continuous frequency variation of:

$$\Delta f_{\text{analog}} = K_{\text{VCO}} \cdot \Delta V_{\text{ctrl}} = 5 \text{ MHz/V} \cdot 0.4 \text{ V} = 2.0 \text{ MHz}$$

The analog MOSCAP was specifically dimensioned to match this exact tuning slope, as verified by the simulated characteristic in *Figure 11*.

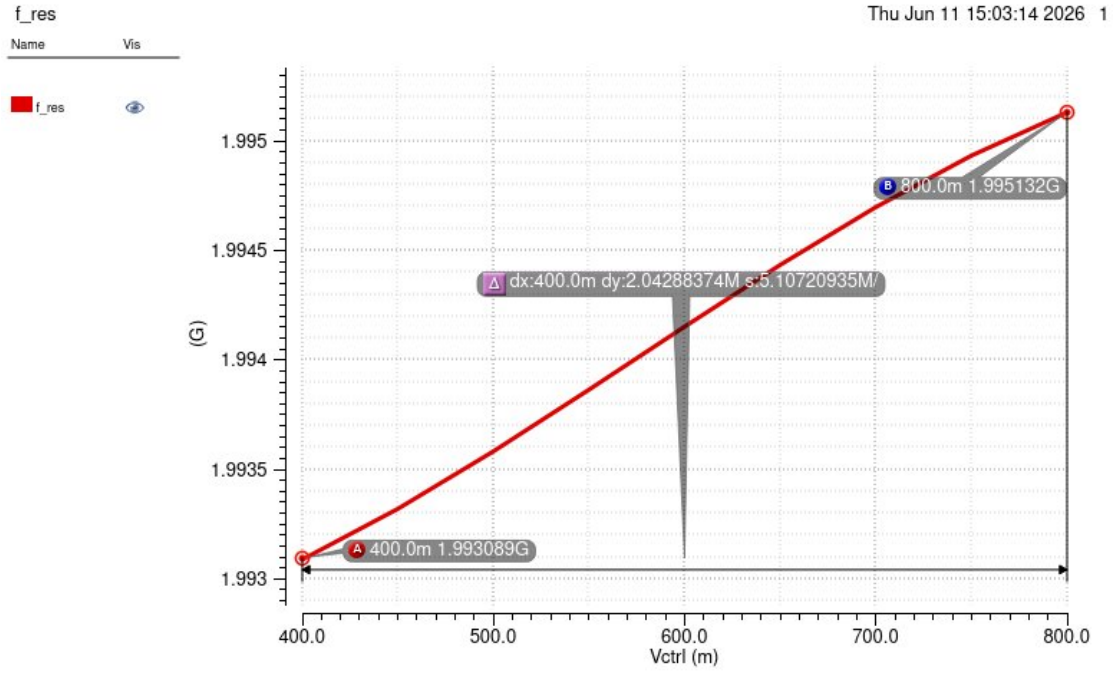


Figure 11: K_{VCO} Gain

Binary-Weighted MOSCAP

To evaluate the varactor's behavior, we set up the testbench shown in *Figure 12* and extracted the effective capacitance using the Z_{11} parameter. As expected, the resulting C-V characteristic closely matches the theoretical model (*Figure 13*).

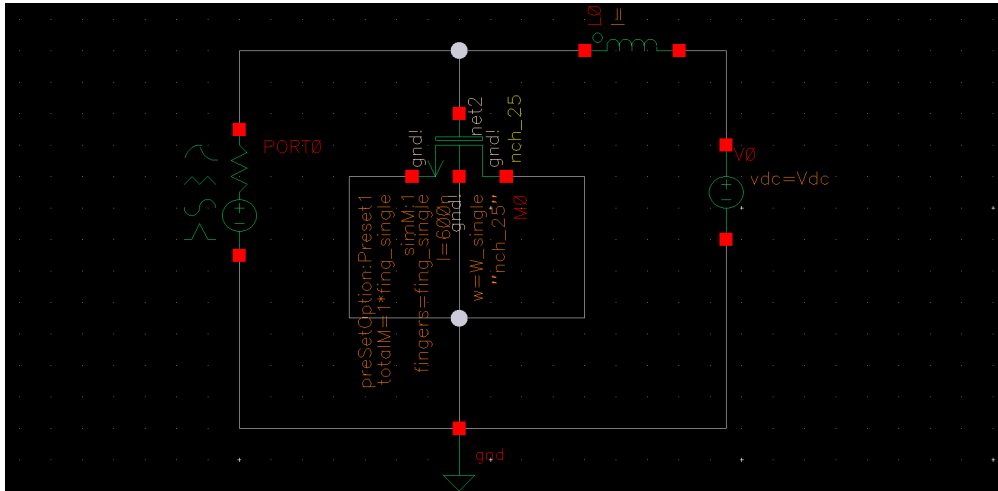


Figure 12: Schematic MOS Varactor

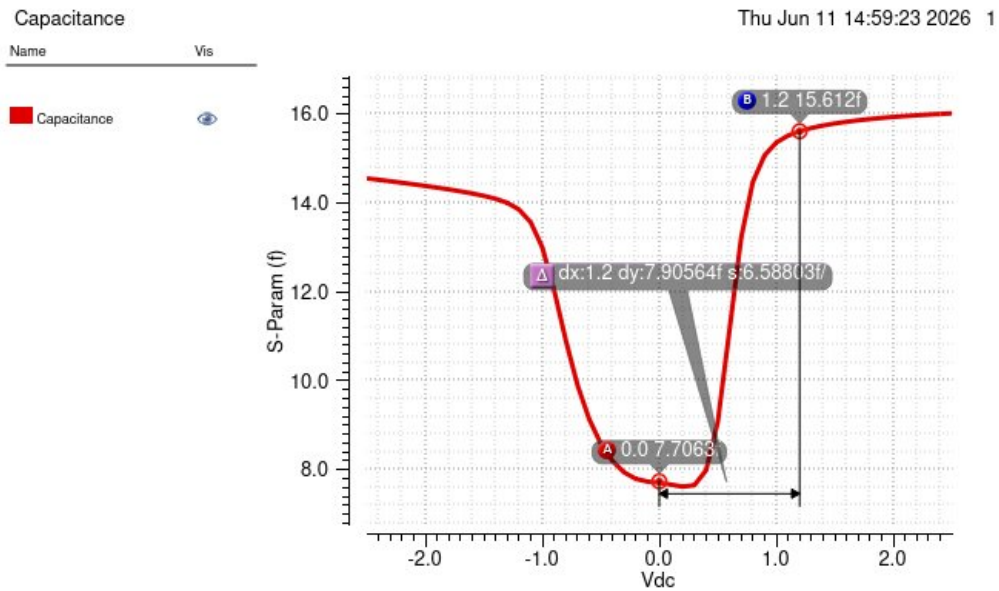


Figure 13: MOS Varactor Characteristic

After confirming the baseline behavior, the device dimensions were optimized to exactly achieve the target unit capacitance step C_0 . In this digitally controlled architecture, the varactor is driven by discrete binary voltage levels (0 V and 1.2 V), restricting it to two distinct capacitance states. Based on the simulation extraction, the device exhibits a minimum capacitance (C_{min}) of approximately 9 fF and a maximum capacitance (C_{max}) of 18 fF. Because C_{min} acts as a permanent parasitic baseline load on the LC tank, the effective capacitance responsible for shifting the resonant frequency is

strictly the differential step (ΔC) between these two states:

$$\Delta C = C_{max} - C_{min} = 18 \text{ fF} - 9 \text{ fF} = 9 \text{ fF}$$

The final sizing of the binary weighted MOSCAP, along with the corresponding design formulas, are detailed in *Figure 14*.

Name	Value	Name	Type	Expr
Filter	Filter	Z11 imOhm	expr	imag(zpm('sp 1 1))
VCO_zTB_single_varactor_1		Capacitance	expr	(-1 / (imag(zpm('sp 1 1)) * 2 * pi * 2e+09))
Analyses			expr	(imag(ypm('sp 1 1)) / (2 * pi * 2e+09))
sp	-2.5 2.5 Automatic Start-Stop 2G			
Design Variables				
Vdc	0			
fing_single	7			
W_single	695n			

Figure 14: Formulas and Sizings of the Binary Weighted MOS Varactor

The binary-weighted MOSCAP shares the same fundamental schematic topology as the analog-controlled varactor, as illustrated in *Figure 15*. However, to achieve the required discrete tuning steps, the device dimensions have been modified. Specifically, the number of fingers has been adjusted, and a CDF parameter is utilized in the instance multiplier to systematically enforce binary scaling across the digital array.

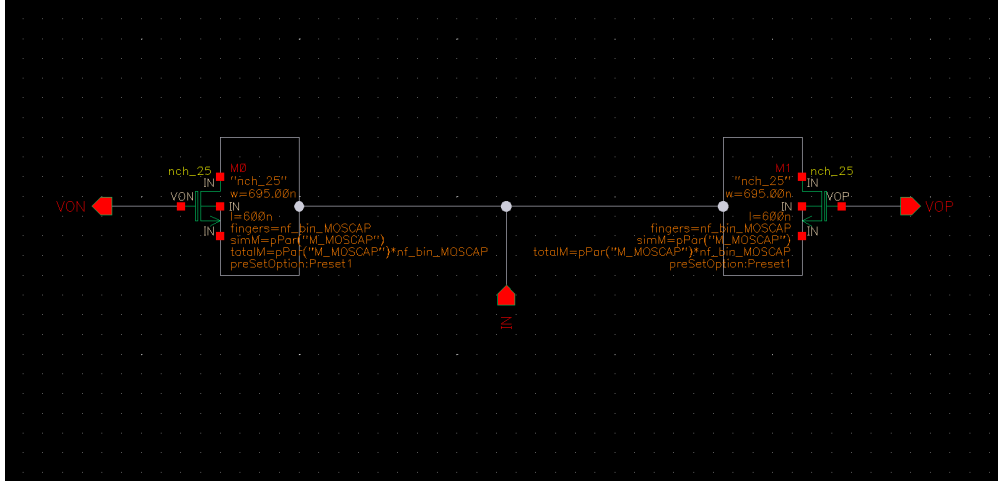


Figure 15: Schematic of the single architecture of the Binary Weighted MOSCAPs

The overall architecture of the binary-weighted MOSCAP array is illustrated in *Figure 16*. To achieve the required discrete capacitance steps, strict binary scaling is enforced across each block. For example, the fourth capacitive instance (controlled by the third bit) utilizes the specific CDF multiplier configuration detailed in *Figure 17*.

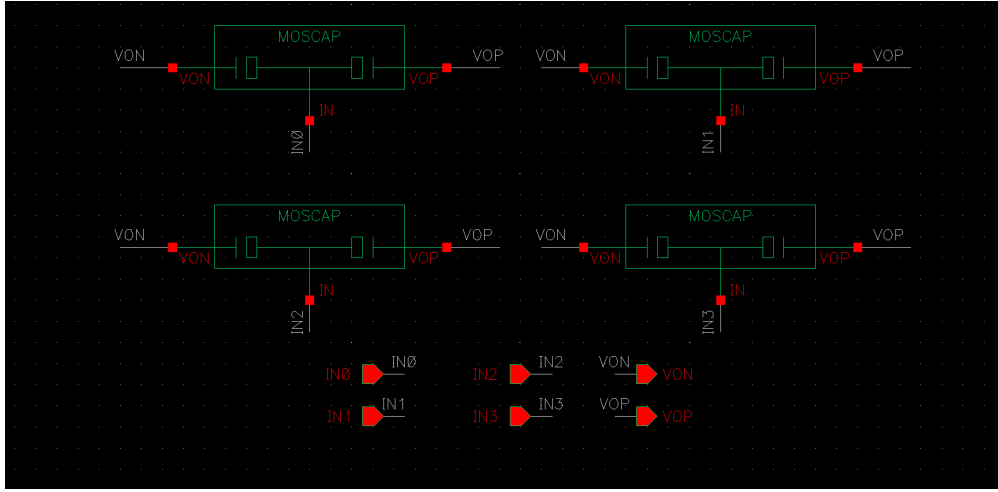


Figure 16: Schematic of the Binary Weighted MOSCAP

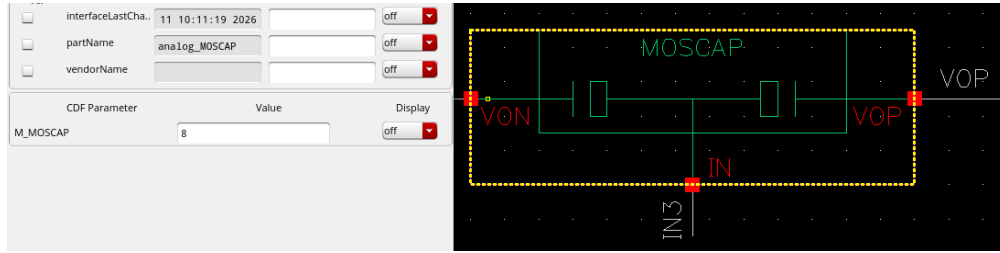


Figure 17: Example of the CDF Parameter of one of the Binary Weighted MOSCAP Instance

Binary-Weighted MIMCAP

In contrast to the fine-tuning MOS varactors, the coarse-tuning bank utilizes a switched-capacitor architecture. Each capacitive branch consists of an NMOS transistor acting as a switch, placed in series between two identical MIM capacitors.

Dimensioning the NMOS switches was one of the most complex aspects of the design, requiring a strict compromise between the quality factor and parasitic loading as discussed in the previous points.

To maintain the correct DC operating points when the switches are OFF, the source and drain nodes are biased through high-value resistors. While an excessively large resistor (e.g., $1\text{ M}\Omega$) provides superior AC isolation to preserve V_{swing} , it injects substantial thermal noise into the tank, degrading phase noise performance. After evaluating the trade-offs among layout area, thermal noise, and voltage swing limits, a resistance of $30\text{ k}\Omega$ was selected as the optimal compromise.

Reaching the target specifications required a rigorous iterative process, simultaneously fine-tuning the switch dimensions, the fixed tank capacitance, the inductor geome-

try, and the MIMCAP unit values. The final optimized sizings that satisfy all constraints are detailed in *Figure 25*.

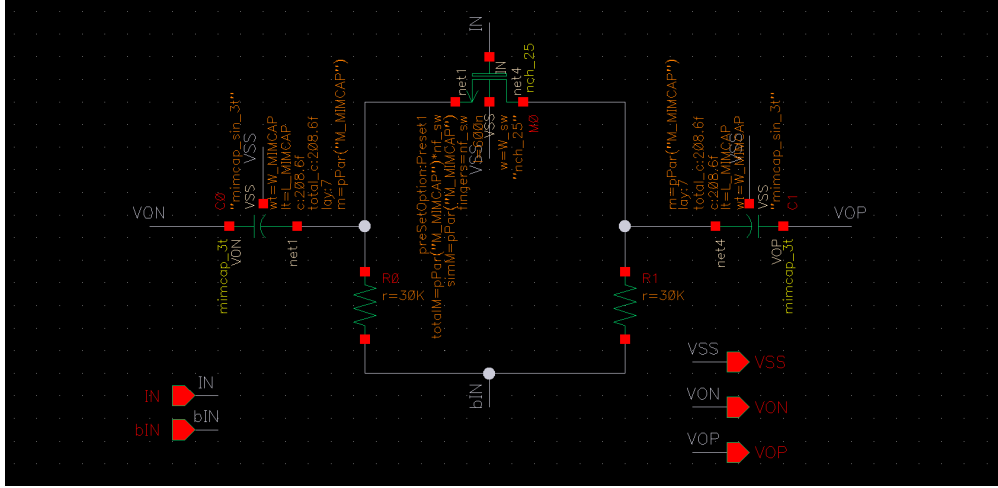


Figure 18: Schematic of the single architecture of the Binary Weighted MIMCAPs

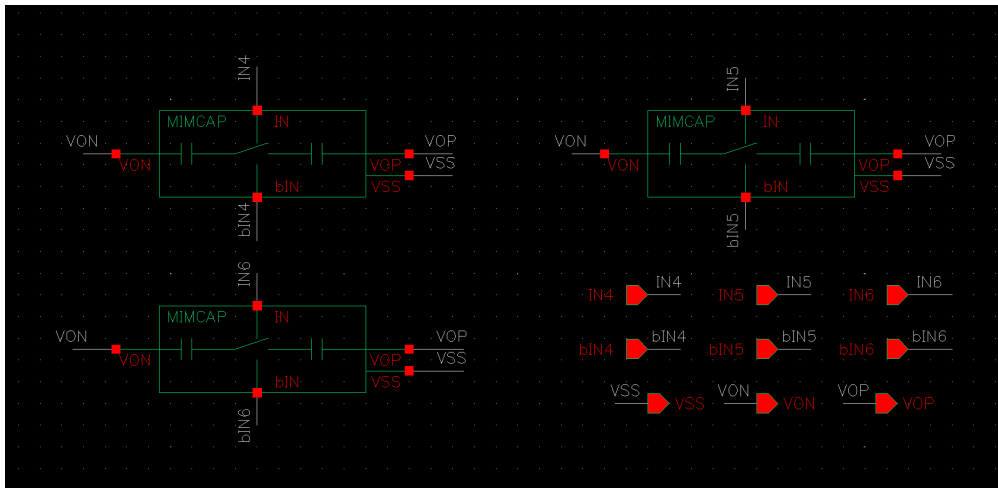


Figure 19: Schematic of the Binary Weighted MIMCAP

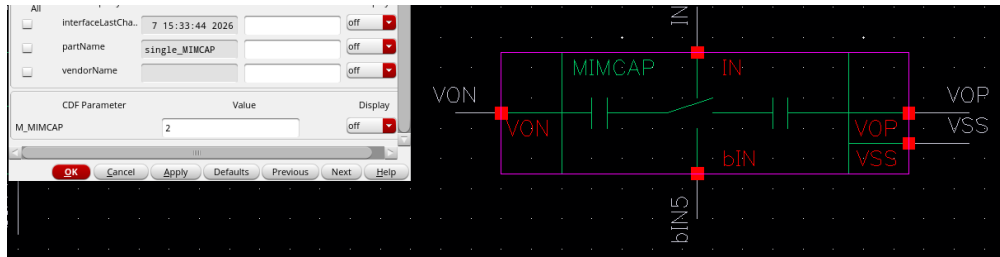


Figure 20: Example of the CDF Parameter of one of the Binary Weighted MIMCAP Instance

Frequency Control Word

The digital capacitor bank is governed by a Frequency Control Word (FCW), generated via a custom Verilog-A decimal-to-binary decoder (as shown in *Figure 21*). The system is designed to enforce a strictly monotonic tuning curve: an FCW of 0 must correspond to the minimum total capacitance (highest frequency), while the maximum FCW must yield the maximum capacitance (lowest frequency).

The Verilog-A logic was explicitly modified to handle the lower four bits (MOSCAPs) differently from the upper bits (MIMCAPs).

In the fine-tuning bank (MOSCAPs), the gate nodes of the MOS varactors are tied directly to the oscillator's common-mode voltage 1.2 V. To reach minimum capacitance (depletion mode, $V_{GS} = 0$), the digital control signal applied to the Source/Drain must be pulled HIGH to 1.2 V. To reach maximum capacitance (strong inversion, $V_{GS} = 1.2$ V), the control signal must be pulled LOW to 0 V. Because the minimum capacitive state requires a HIGH control voltage, the Verilog-A outputs for the first four bits had to be inverted; without this critical modification, the highest frequency state would have incorrectly occurred at a decimal value of 15 rather than 0. Conversely, the coarse-tuning bank (MIMCAPs) representing the upper bits of the Verilog-A module required no inversion. The resulting FCW correctly guarantees that any integer increase uniformly injects more capacitance into the tank, smoothly lowering the resonant frequency.

```

1 // VerilogA for VCO, FCWgen, veriloga
2
3 `include "constants.vams"
4 `include "disciplines.vams"
5
6 (* ignore_hidden_state *)
7 module dec2bin_ClassD(Vdd_bCTRL, Vdd_CTRL, Vss,
8 CTRL, bCTRL);
9 input Vdd_bCTRL, Vdd_CTRL, Vss;
10 electrical Vdd_bCTRL, Vdd_CTRL, Vss;
11
12 output [0:6] CTRL;
13 electrical [0:6] CTRL;
14 output [0:6] bCTRL;
15 electrical [0:6] bCTRL;
16
17 parameter integer inval = 0;
18 real tr = 100f;
19 integer res, next;
20 real CTRL_sig[0:6], bCTRL_sig[0:6];
21 genvar i, j;
22
23 analog begin
24 @(initial_step) begin
25 res = 0;
26 next = inval;
27 for(i=0; i<7; i=i+1)begin
28 res = next % 2;
29 next = (next-res) / 2;
30
31 if (i<=3) begin
32 CTRL_sig[i] = (1-res)*V(Vdd_CTRL);
33 bCTRL_sig[i] = res*V(Vdd_bCTRL);
34 end else begin
35 CTRL_sig[i] = res*V(Vdd_CTRL);
36 bCTRL_sig[i] = (1-res)*V(Vdd_bCTRL);
37 end
38 end
39 end
40 for(i=0; i<7; i=i+1) begin
41 V(bCTRL[i],Vss) <+ transition(bCTRL_sig[i], 0, tr, tr);
42 end
43 for(j=0; j<7; j=j+1) begin
44 V(CTRL[j],Vss) <+ transition(CTRL_sig[j], 0, tr, tr);
45 end
46 end
47 endmodule

```

Figure 21: Final VerilogA Script

Final Design

The complete top-level schematic of the digitally controlled LC Voltage Controlled Oscillator is illustrated in *Figure 22*. This top-level schematic integrates all sub-blocks, illustrating the parallel connection of the active cross-coupled core, the LC tank, and the complete hybrid tuning network. It demonstrates how the continuous analog varactor operates alongside the 8-bit digital array (fine MOSCAP and coarse MIMCAP banks) to achieve the target frequency range.

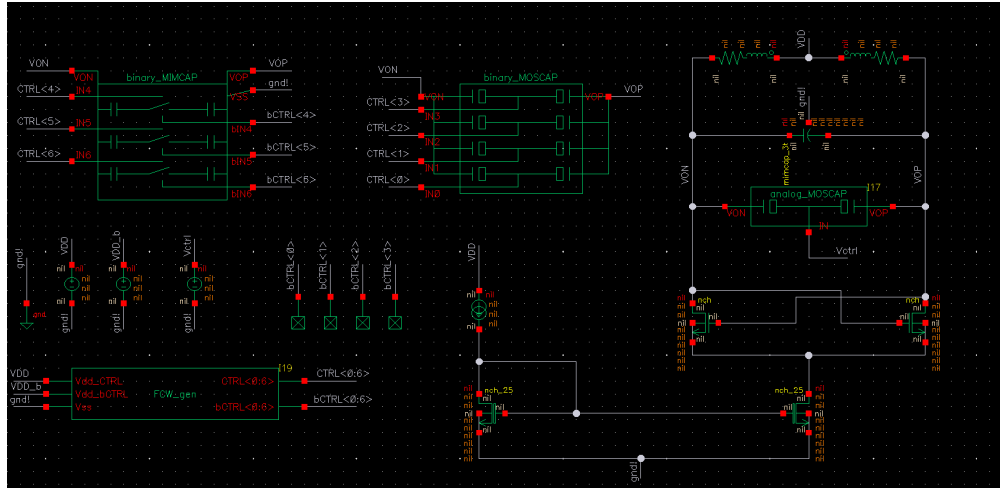


Figure 22: Schematic of the complete LC VCO

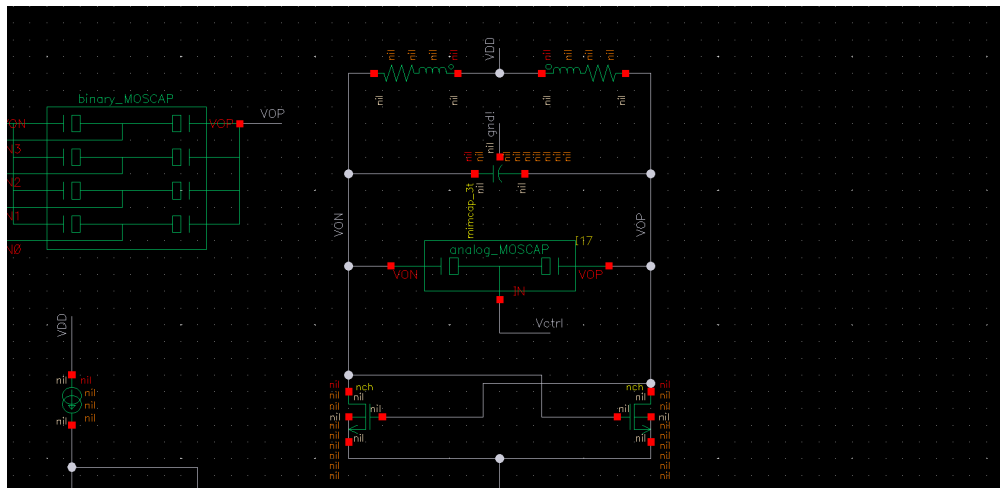


Figure 23: Zoom over the XCP and fixed L and C

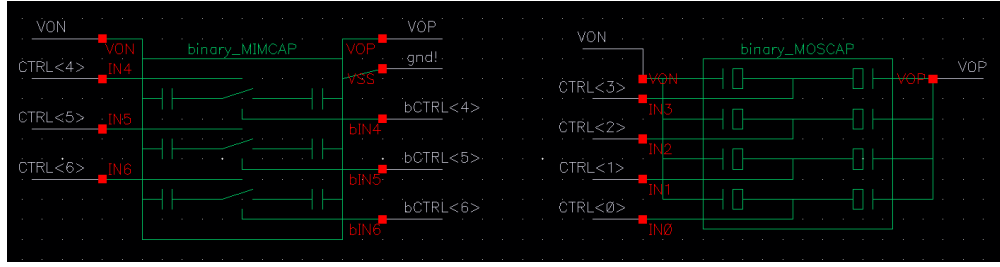


Figure 24: Zoom over Digital Capacitor Banks

As detailed in *Figure 25*, the final dimensional parameters of the analog-controlled MOSCAP (`nf_varactor`) and the binary-weighted MOSCAP array (`nf_bin_MOSCAP`) are strictly constrained by the target K_{VCO} gain and the chosen unit capacitance (C_0), respectively. Furthermore, `nf_sw` and `W_sw` denote the number of fingers and the channel width of the NMOS switch transistors employed within the MIMCAP coarse-tuning array.

The fixed tank capacitor (`Fixcap`), sized with these optimized dimensions, contributes a baseline capacitance of approximately 1.78 pF. Notably, this fixed value is intentionally smaller than the initially predicted C_{min} of the tank. This reduction is required to accurately compensate for the non-zero capacitance inherently injected by the tunable MOSCAP and MIMCAP branches, even when their corresponding digital control bits are LOW.

VCO_ClassB_VCO_1	
Simulator spectre	
Analyses	
☒ tran	0 100n conservative
☒ dc	t
☒ pss	2G 100 /VOP /VON
☒ pnoise	100 1k 2G/2 /VOP /VON
Click to add analysis	
Design Variables	
☒ nf_varactor	12
☒ nf_bin_MOSCAP	7
☒ val_verilogA	64
☒ Vctrl	600m
☒ nf_sw	10
☒ W_sw	17u
☒ L_fixcap	W_fixcap
☒ W_fixcap	29.58u
☒ L_MIMCAP	W_MIMCAP
☒ W_MIMCAP	10.3u
☒ L_ind	1.3n

Figure 25: Final Sizing of the Design

Figure 26 summarizes the primary performance metrics of the oscillator evaluated under nominal operating conditions. Driven by a fixed tail current of $I_{SS} = 2\text{ mA}$, the cross-coupled core successfully establishes steady-state resonance at the target center frequency of 2.0 GHz, while sustaining a single-sided voltage swing strictly greater than the 0.5 V requirement. Furthermore, the phase noise performance is optimized, yielding a Figure of Merit (FoM) of -187 dBc/Hz at a 100 kHz frequency offset.

v /VOP; tran (V)	expr	vtime(tran "VOP")	
v /VON; tran (V)	expr	vtime(tran "VON")	
v /VOP; pss (V)	expr	vtime(pss "VOP")	
v /VON; pss (V)	expr	vtime(pss "VON")	
OUT_Diff	expr	(vtime(pss "VOP") - vtime(pss "VON"))	
i /V0/MINUS Vctrl=400m; pss (I)	expr	itime(pss "V0/MINUS")	
Power	expr	abs((1.2 * average(itime(pss "V0/MINUS"))))	
noise_100k	expr	value(pn("pnoise") 100000)	
ISS	expr	IDC("M2/D")	2.006m
peak_OUT_Diff	expr	ymax(vtime(pss "VOP") - vtime(pss "VON"))	539.3m
f_res	expr	harmonicFreq((v("VOP") ?result "pss_fd") - v("VON") ?result "pss_fd") 1)	2G
FoM_100k	expr	((noise_100k - dB20(f_res / 100000)) + dB10(Power / 0.001))	-187

Figure 26: Oscillator Performance Metrics

The frequency tuning capabilities of the architecture are validated in Figure 27. The combined action of the analog varactor and the 7-bit digital array provides a continuous tuning range that safely covers the required spectrum from 1.9 GHz to 2.1 GHz.

Parameters: val_verilogA=0					
1	VCO_ClassB_VC...	ISS	2.006m		
1	VCO_ClassB_VC...	peak_OUT_Diff	711.6m		
1	VCO_ClassB_VC...	f_res	2.132G		
Parameters: val_verilogA=64					
2	VCO_ClassB_VC...	ISS	2.006m		
2	VCO_ClassB_VC...	peak_OUT_Diff	539.3m		
2	VCO_ClassB_VC...	f_res	2G		
Parameters: val_verilogA=127					
3	VCO_ClassB_VC...	ISS	2.006m		
3	VCO_ClassB_VC...	peak_OUT_Diff	396.8m		
3	VCO_ClassB_VC...	f_res	1.89G		

Figure 27: Tuning Range Results

To verify the continuous frequency coverage of the digitally controlled oscillator, the tuning range across the entire FCW spectrum has been simulated and divided into four segments for clarity. The following results explicitly demonstrate that the tuning range has no holes and maintains sufficient bank overlap when passing from one discrete capacitive bank to the other.

To definitively prove the absence of blind zones, arbitrary horizontal lines representing constant target frequencies have been superimposed across the plots. As visible in the detailed overlap figures, these horizontal markers consistently intersect multiple adjacent tuning curves, ensuring that the PLL can lock onto any arbitrary frequency within the operational band without encountering dead zones.

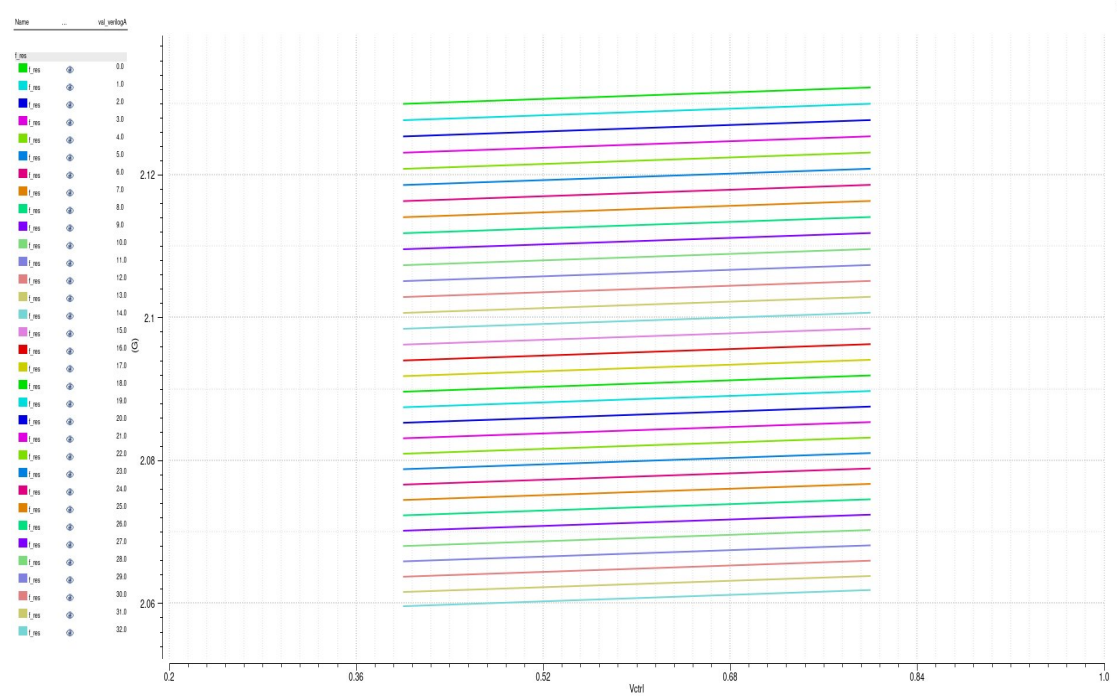


Figure 28: Simulated tuning range for FCW codes 0 to 32

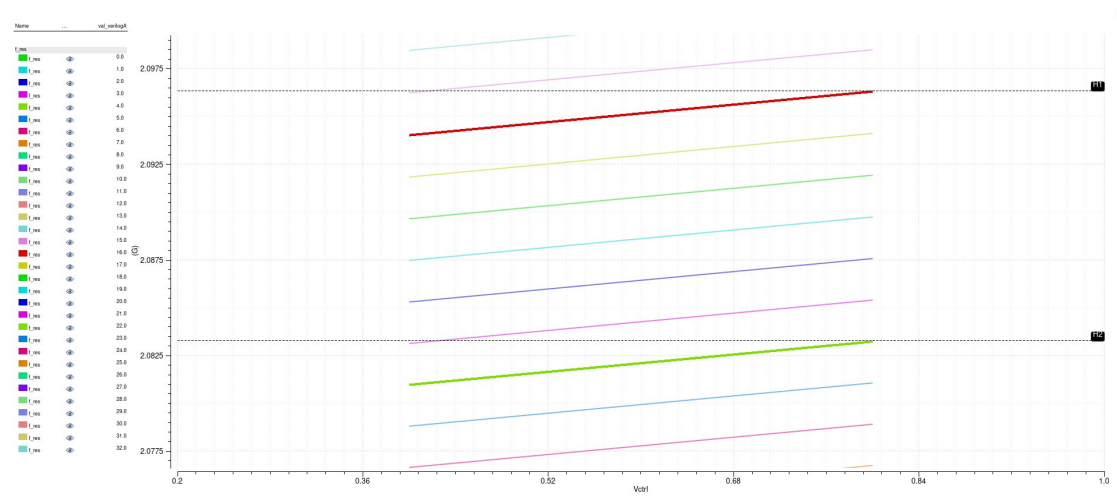


Figure 29: View of the capacitive bank overlap for codes 0 to 32

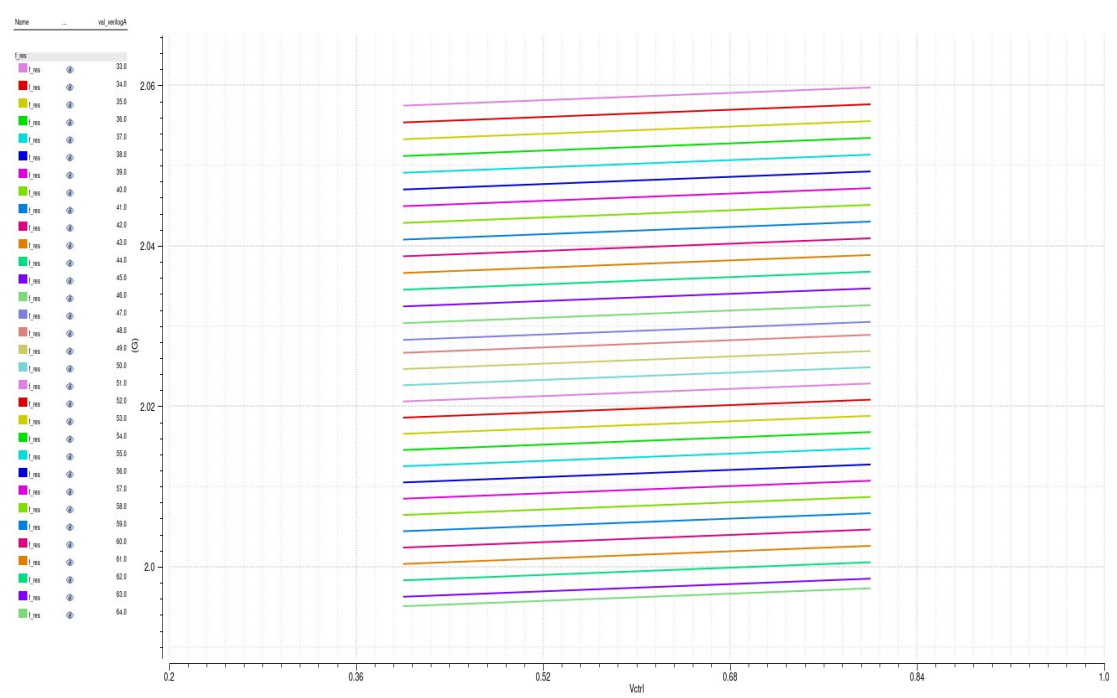


Figure 30: Simulated tuning range for FCW codes 32 to 64

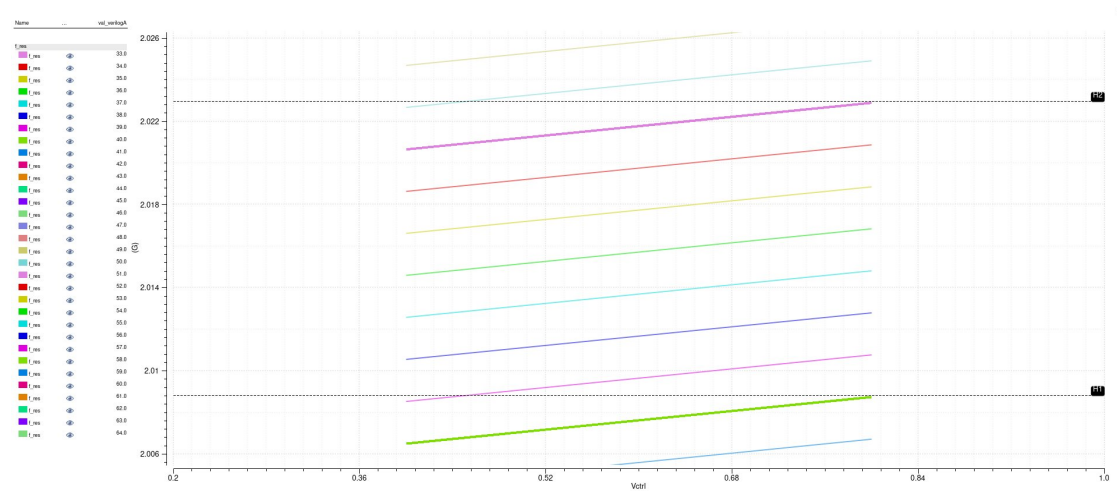


Figure 31: View of the capacitive bank overlap for codes 32 to 64

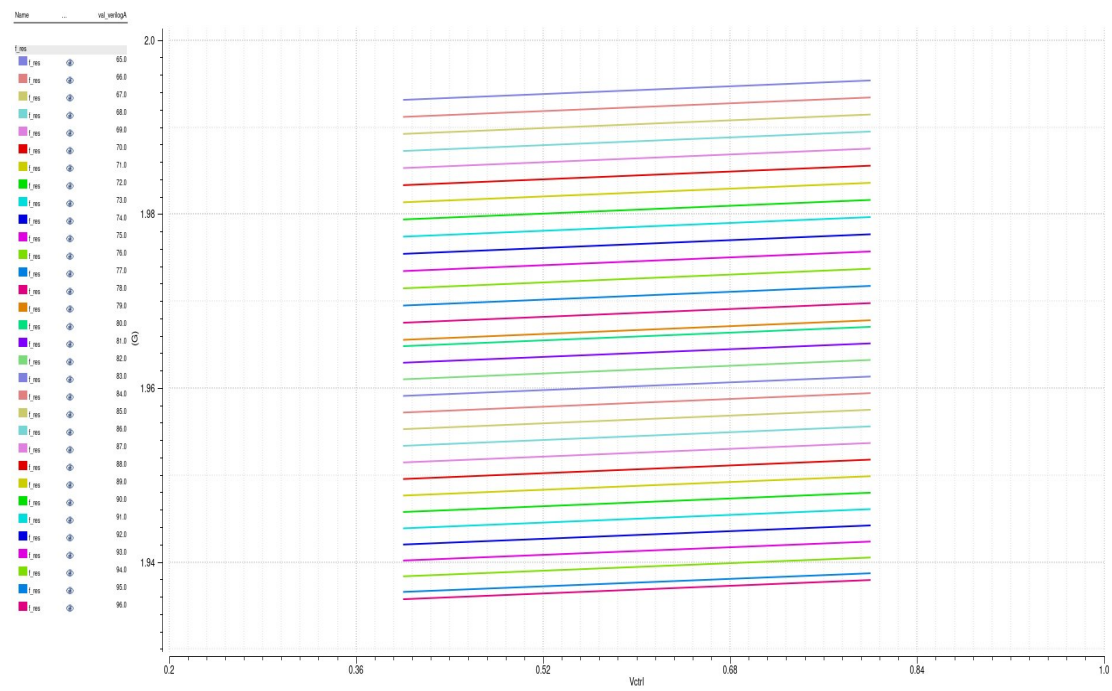


Figure 32: Simulated tuning range for FCW codes 64 to 96

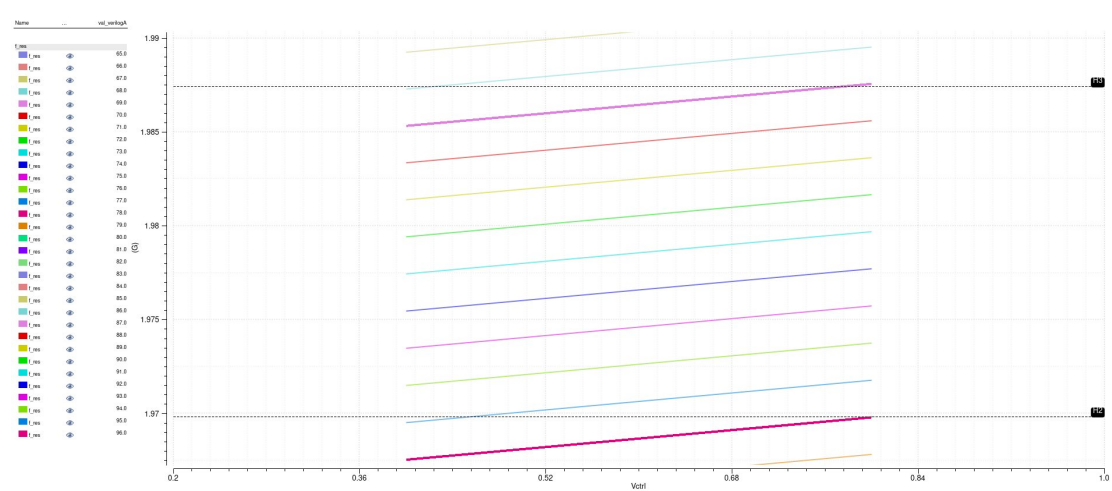


Figure 33: View of the capacitive bank overlap for codes 64 to 96

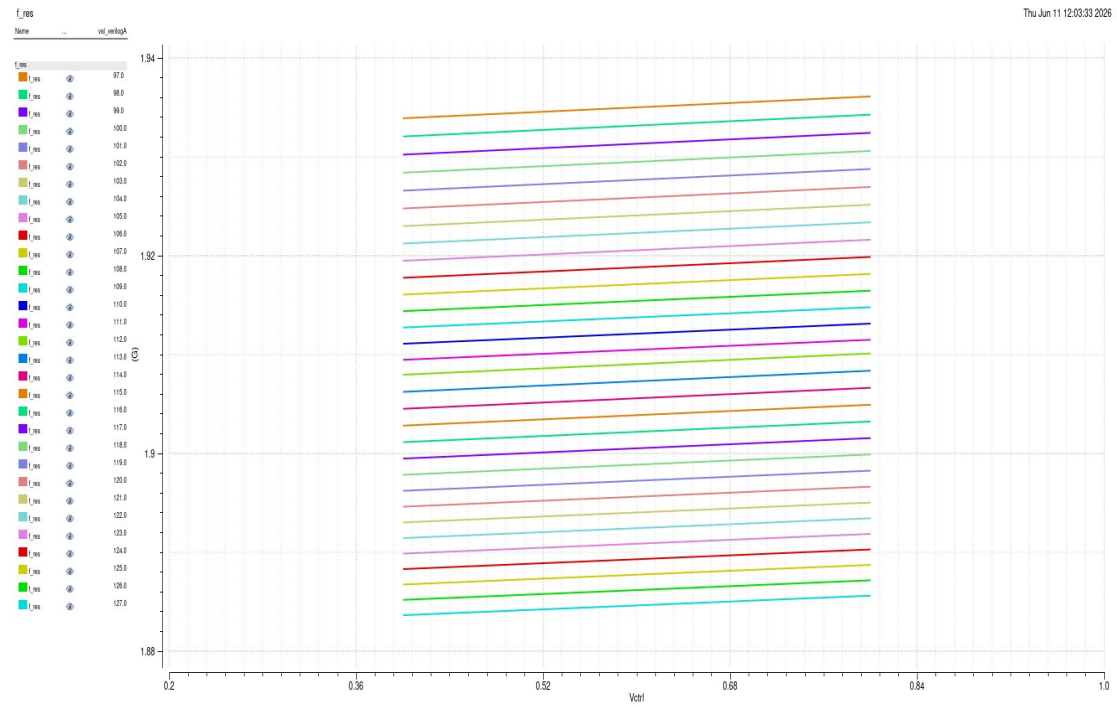


Figure 34: Simulated tuning range for FCW codes 96 to 127

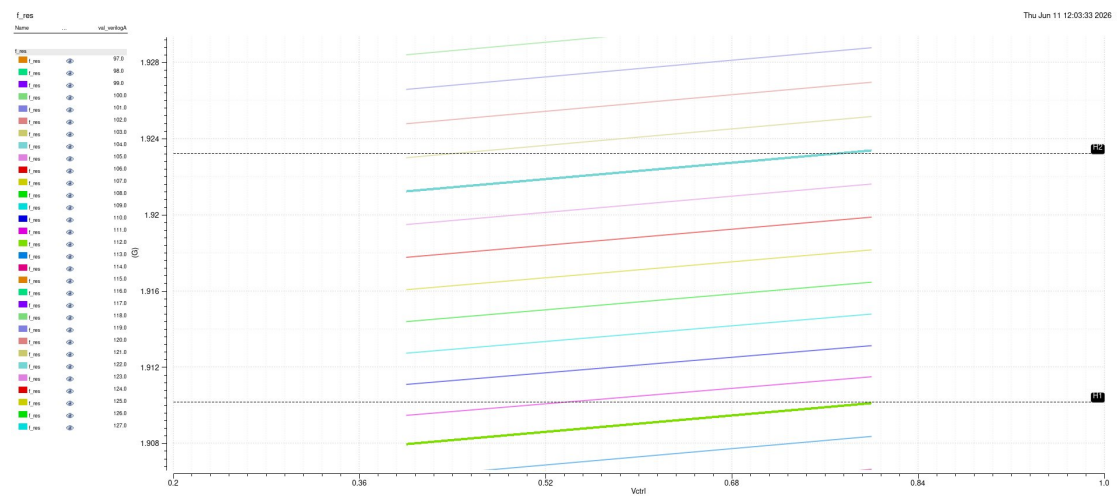


Figure 35: View of the capacitive bank overlap for codes 96 to 127